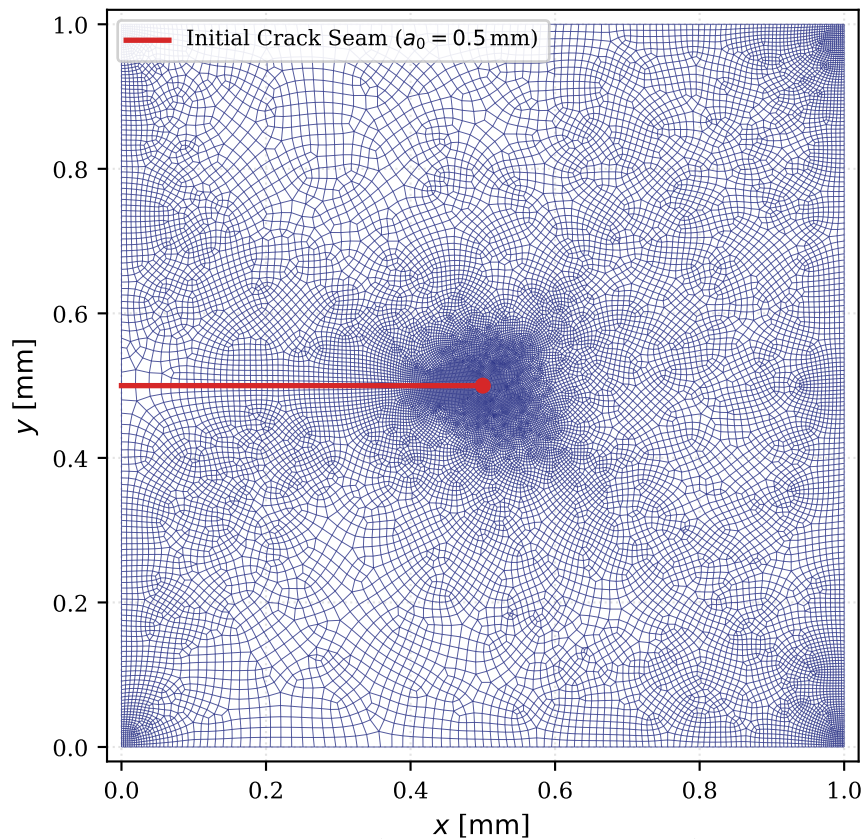
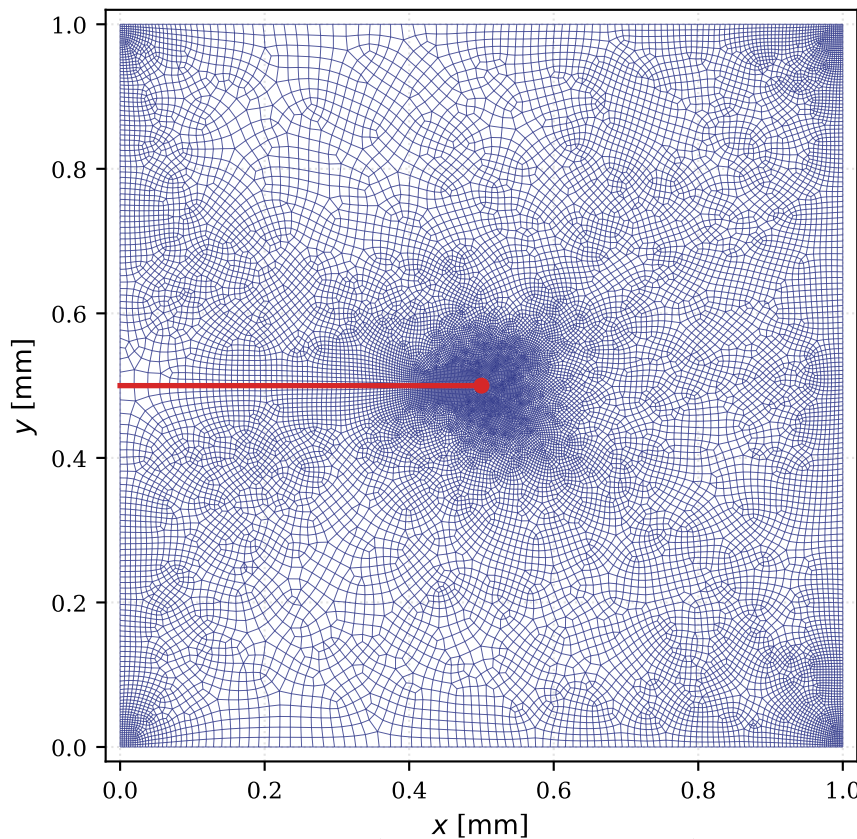


Mode-II Native Adaptive Remeshing: Three-Way Discretization Comparison
(Source: Job-1_UEL_paper_horizon.odb, Rule: UNIFORM_ERROR, Variables: MISESeri)

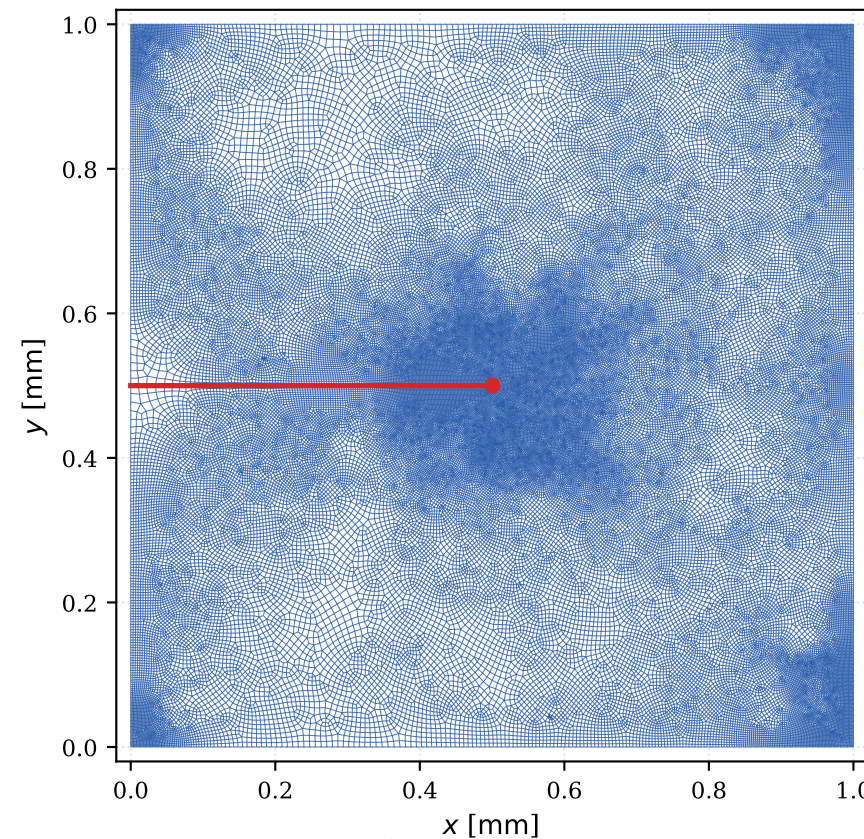
Step-1 ($u_x = 10\text{ }\mu\text{m}$, ET = 2.0%)
Full Domain: 22,530 FEs, 22,642 nodes



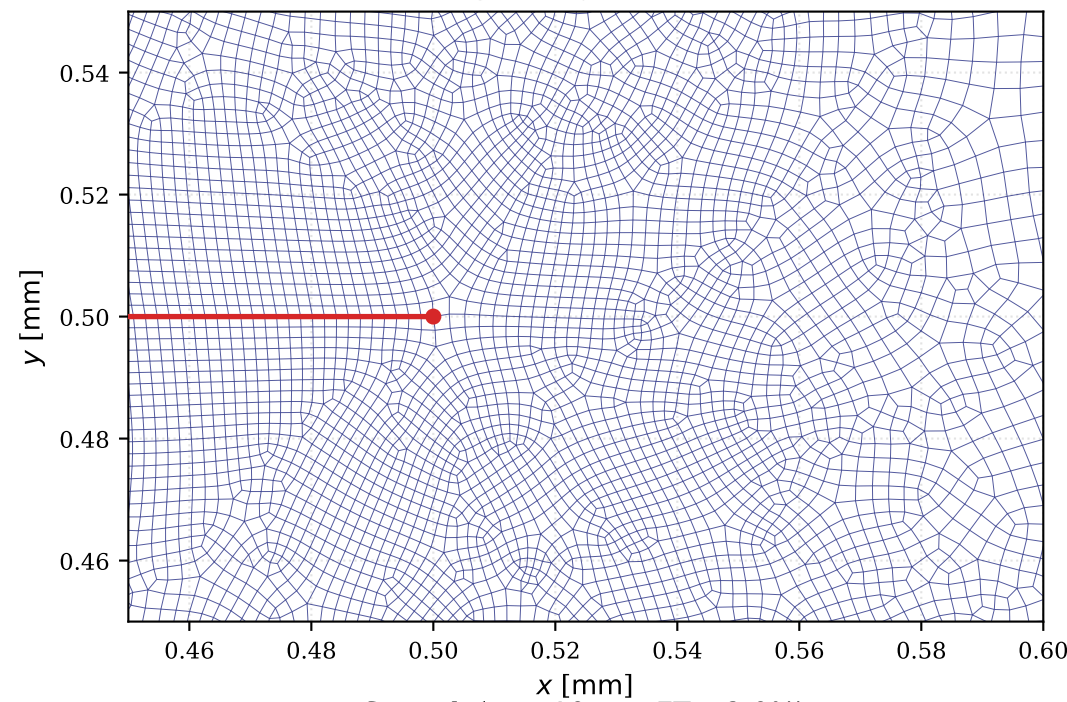
Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 2.0%)
Full Domain: 22,405 FEs, 22,512 nodes



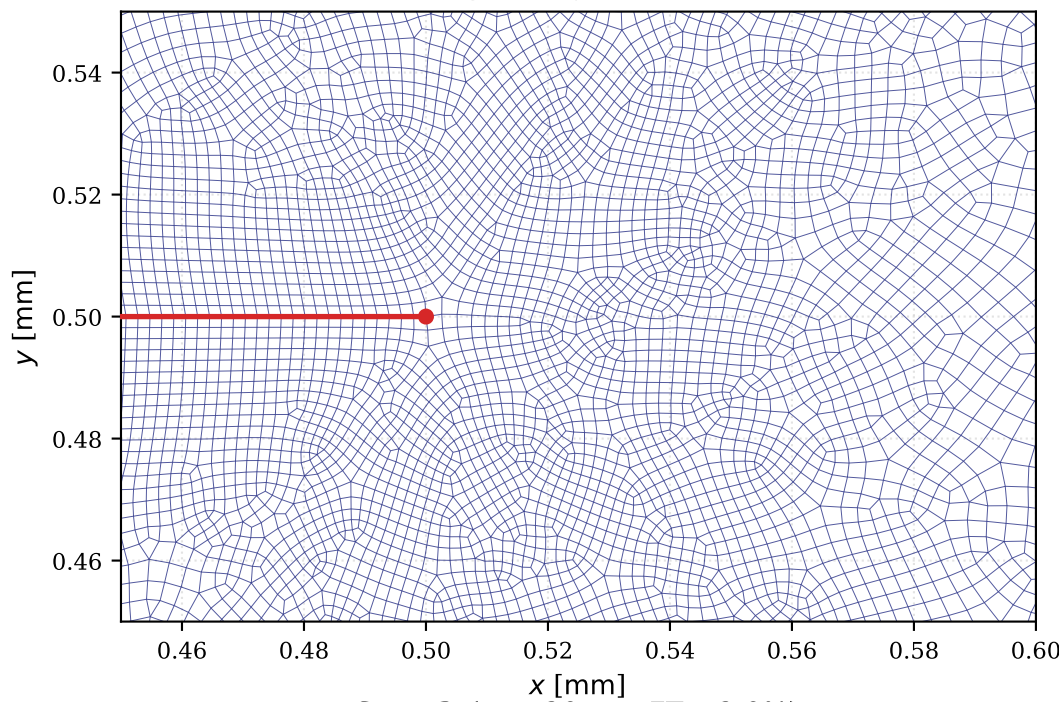
Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 1.0%)
Full Domain: 80,474 FEs, 80,136 nodes



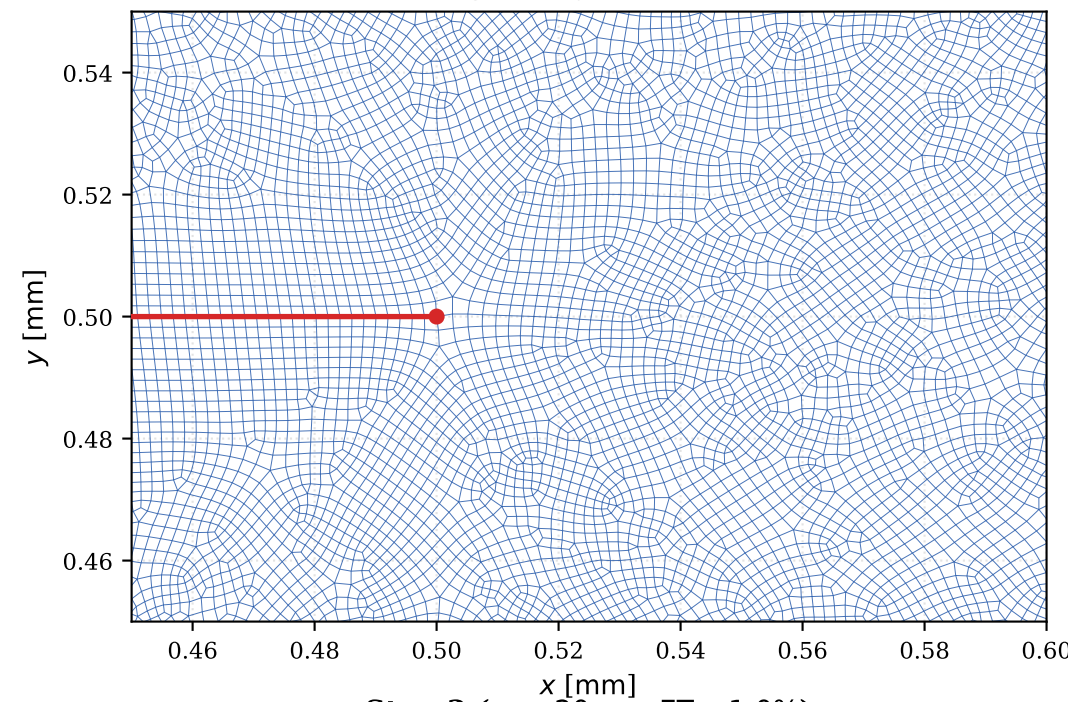
Step-1 ($u_x = 10\text{ }\mu\text{m}$, ET = 2.0%)
Crack-Tip Singularity Zoom [$h_{\min} = 1.0\text{ }\mu\text{m}$]



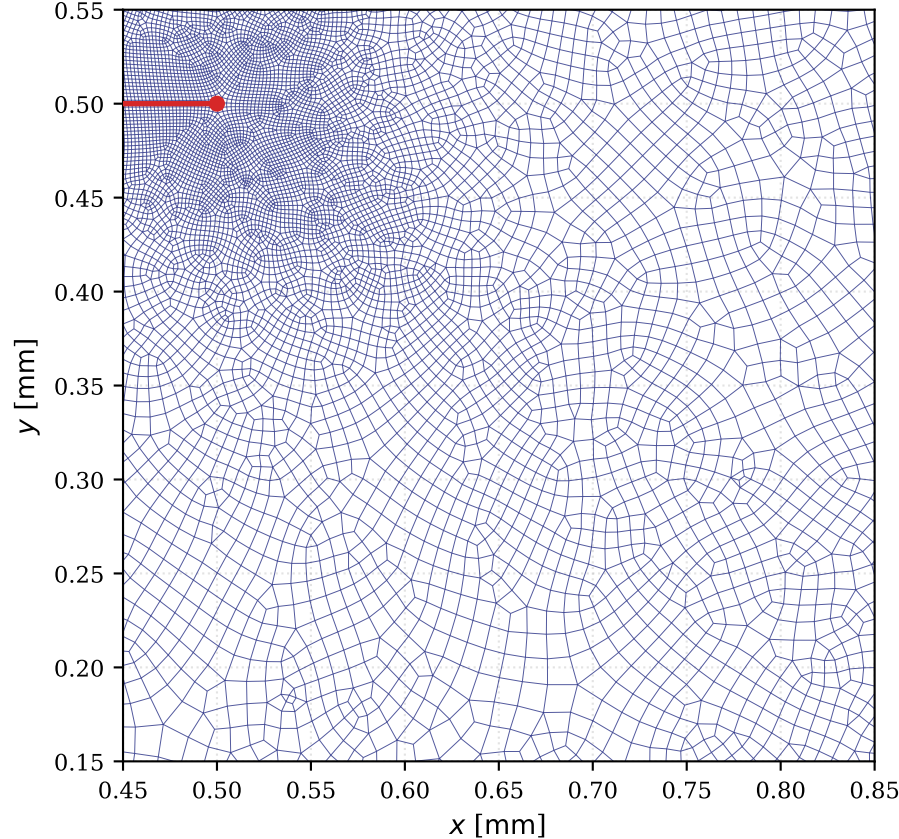
Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 2.0%)
Crack-Tip Singularity Zoom [$h_{\min} = 1.0\text{ }\mu\text{m}$]



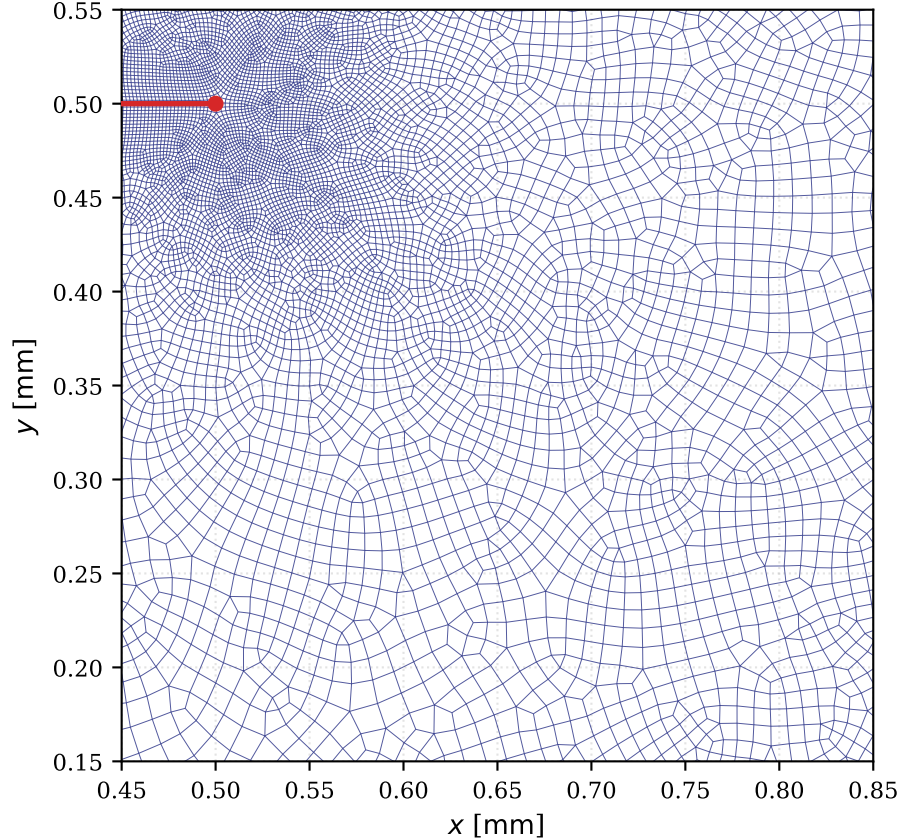
Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 1.0%)
Crack-Tip Singularity Zoom [$h_{\min} = 0.60\text{ }\mu\text{m}$]



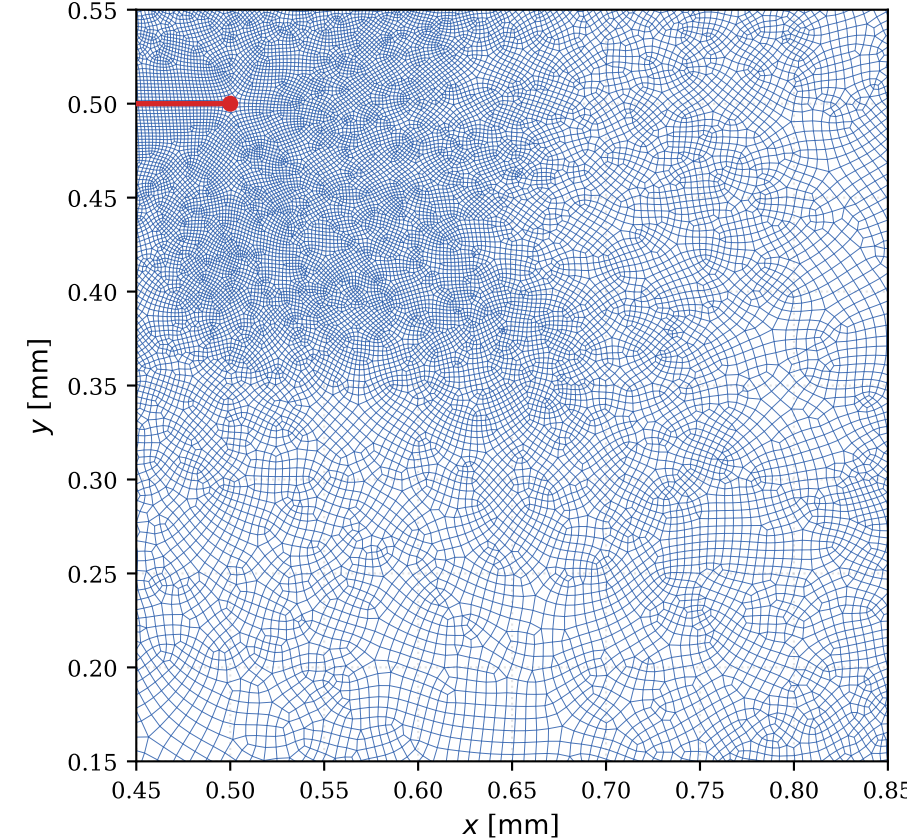
Step-1 ($u_x = 10\text{ }\mu\text{m}$, ET = 2.0%)
Refinement Corridor & Shear Fan



Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 2.0%)
Refinement Corridor & Shear Fan



Step-2 ($u_x = 20\text{ }\mu\text{m}$, ET = 1.0%)
Extended Corridor (+259% FE Density)



QUANTITATIVE MESH TOPOLOGY & PROVENANCE SUMMARY:

- Step-1 ET=2.0% Mesh (Retest 1411103): 22,530 FEs (21,962 quads + 568 tris, 22,642 nodes) | SHA-256: bd02d73c2bc199db...
- Step-2 ET=2.0% Mesh (Candidate) : 22,405 FEs (21,827 quads + 578 tris, 22,512 nodes) | SHA-256: 9c453eb3b004c3a0...
- Step-2 ET=1.0% Mesh (Diagnostic) : 80,474 FEs (78,363 quads + 2,111 tris, 80,136 nodes) | SHA-256: 899528651ba68b50...
- Topological Insights: Step-1 vs Step-2 at ET=2.0% are scale-invariant (-0.55% FE delta). At ET=1.0%, element count expands by +259.2%, extending the fine-mesh envelope ($h \leq 8\text{ }\mu\text{m}$ for 99.3% of domain) deeply along the theoretical shear path towards (0.81, 0.0) mm.